

Experiment Number:02

Experiment Name: Simplify functions & implement them by logic gates

Theory:

Karnaugh Map (K-Map) is a graphical method used to simplify Boolean expressions. It arranges the truth-table values in Gray code order, allowing adjacent 1s to be grouped together. By forming groups of 1, 2, 4, 8, etc., redundant variables can be eliminated and a simpler Boolean expression can be obtained. This reduces the number of logic gates required for circuit implementation.

After simplifying the Boolean function using a K-Map, the resulting expression can be implemented using logic gates such as AND, OR, and NOT gates. Each product term is implemented using AND gates, complemented variables are generated using NOT gates, and the product terms are combined using an OR gate. Thus, K-Map simplification helps design a circuit that is simpler, faster, and uses fewer hardware components.

Functions:

$$1. F(A,B,C) = A'BC + AB'C + ABC' + ABC$$

$$2. F(A,B,C) = A(A+B)(A+B+C)$$

$$3. F(A,B,C) = (A + (BC)')'(AB + ABC)$$

$$4. F(A,B,C) = (B'(A+B) + (A+B)(A+B'))B'$$

Implementations:

$$1. F(A,B,C) = A'BC + AB'C + ABC' + ABC$$

K-MAP:

A\BC	B'C'	B'C	BC	BC'
A'	0	0	1	0
A	0	1	1	1

Simplified version: $AB + BC + CA$

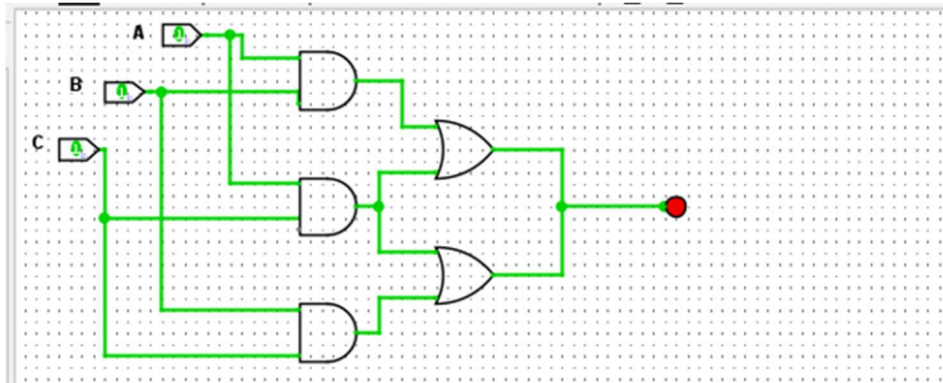


Fig 01: Gate implementation of function 1

$$2. F(A,B,C) = A(A+B)(A+B+C)$$

K-MAP:

C\AB	A'B'	A'B	AB	AB'
C'	0	0	1	1
C	0	0	1	1

Simplified version: A

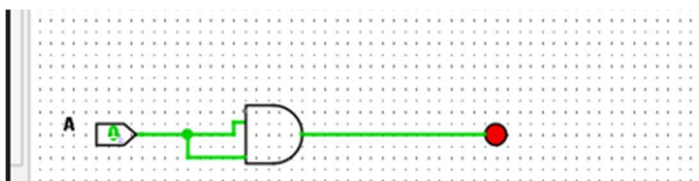


Fig 02: Gate implementation of function 2

$$3. F(A,B,C) = (A + (BC)')'(AB + ABC)$$

K-MAP:

C\AB	A'B'	A'B	AB	AB'
C'	0	0	0	0

C	0	0	0	0
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Simplified version:0

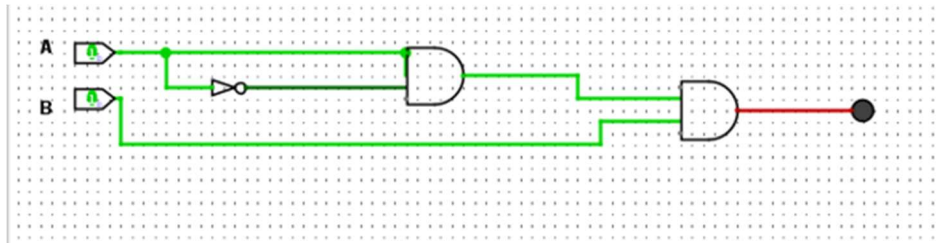


Fig 03: Gate implementation of function 3

$$4.F(A,B,C)=(B'(A+B)+(A+B)(A+B'))B'$$

K-MAP:

C\AB	A'B'	A'B	AB	AB'
C'	0	0	0	1
C	0	0	0	1

Simplified version: AB'

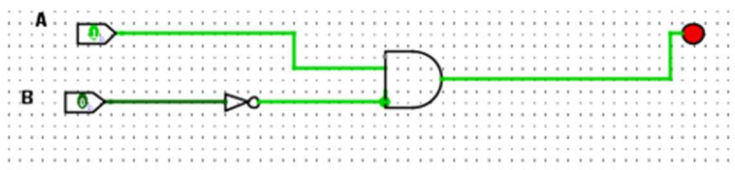


Fig 04: Gate implementation of function 4